A MILP-Based Decision Support System for Capacity Allocation in Supplier-Managed Inventory

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Abstract. This study proposes a mixed-integer linear programming (MILP) model to address the single-supplier, multi-buyer supplier-managed inventory problem by optimizing order allocation under capacity and inventory constraints. A decision support system (DSS) is developed to integrate enterprise data, model instantiation, and automated solution generation within an interactive framework. The approach is validated through a real-world case study in the automotive industry, demonstrating that the model can be solved efficiently on an operational scale and produces stable and reasonable allocation decisions. The results indicate that total system cost is predominantly driven by backorder costs, with the model systematically allocating capacity to mitigate backlog accumulation while maintaining minimal constraint violations. The proposed framework provides a structured foundation for integrating optimization into practical supply chain planning and offers flexibility for future extensions.

Keywords: Decision Support System, Supplier-Managed Inventory, Mixed Integer Linear Programming, Supply Chain Optimization, Automotive Industry

1 Introduction

Vertical integration has been recognized as a fundamental corporate strategy in the fields of strategic management and organizational economics [1]. In the context of supply chain management, vertical integration refers to a firm's ownership and control over multiple stages of the value chain, therefore, can be considered an attractive approach to supply chain design and reconfiguration [2]. As firms redesign and reconfigure their supply chain in increasing environments, vertical integration is often viewed as a structural mechanism to enhance responsiveness and operational alignment. However, firms implementing vertical integration must determine whether to focus on upstream activities, such as raw materials and component manufacturing, or downstream activities, such as assembly and distribution, since each configuration requires different coordination mechanism and information infrastructures [3].

From an upstream perspective, vendor-managed inventory (VMI) represents a widely adopted coordination mechanism that shifts inventory control toward upstream stages of the supply chain [4]. VMI is a partnering initiative aimed at enhancing efficiency across supply chains by allowing the supplier to monitor the buyer's inventory

and make replenishment decisions regarding order quantity, timing, and delivery [5, 6]. Under the VMI program, the suppliers' responsibility in managing inventory has been emphasized [7]. Therefore, effective VMI implementation requires the sharing of information among supply chain partners [8]. Based on VMI principles, supplier-managed inventory (SMI) is a similar approach but focuses on the inbound flow to manufacturing, where material replenishment is triggered by the manufacturer's production schedule, rather than restocking finished goods [9]. To the best of our knowledge, SMI has received considerably less attention than VMI in existing literature. Optimization-oriented SMI model remains relatively limited and underdeveloped compared to the extensive research on VMI.

This study investigates a single-supplier, multi-buyer supplier-managed inventory problem (SS-MB-SMI) through a case study in an automotive supply chain. In this context, the upstream supplier produces and provides component sets for car seat covers that are delivered to multiple downstream vehicle assembly plants within the same corporate group, reflecting a vertical integration strategy within the supply chain. The vertically integrated structure enables centralized monitoring of plant-level inventory data and coordinated allocation of limited production capacity across plants. This operational setting requires a computational decision framework capable of balancing inventory risks, production constraints and logistics efficiency in a multi planning horizon.

The contributions of this study are positioned at the intersection of operations research and information technology for decision support. First, a capacitated multi-buyer SMI optimization model is developed, incorporating production capacity limits, case-pack constraints, and inventory balancing mechanisms. Second, the problem is formulated as a mixed-integer linear programming (MILP) model suitable for implementation within enterprise-level planning systems. Third, a decision support system (DSS) framework is designed to operationalize the model, enabling automated order allocation, real-time inventory visibility, and coordinated decision-making across organizational units. This integration of mathematical modeling and DSS architecture contributes to IT-enabled supply chain coordination under vertical integration.

The reminder of the paper is structured as follows: Section 2 reviews the relevant literature and clarifies the research contribution. Section 3 presents the mathematical formulation of the SS-MB-SMI model. Section 4 describes the decision support system framework. Section 5 reports the results and evaluation. Finally, Section 6 concludes the study by summarizing the main findings and directions for future research.

2 Related Work

VMI has attracted considerable attention as an effective mechanism for coordinating supply chain activities of single-vendor multi-buyer models [10]. Although SMI is closely related to VMI, mathematical models explicitly developed for SMI remain scarce. To clarify this gap, Table 1 reviews representative VMI-based studies and highlights their main modelling characteristics and solution approaches, providing a reference framework for positioning the resent study.

Research on single-vendor multi-buyer inventory systems has been developed under the VMI paradigm, with the primary objective of improving coordination between vendors and buyers through optimized replenishment decisions. Early studies focused on deterministic demand settings to demonstrate the benefits of centralized optimization over decentralized decision-making [11, 12]. Subsequent research improved their models by considering more realistic operational factors, including multiple products, production capacity constraints, and alternative stockout responses such as backorder and lost sales [13, 14, 15]. Despite these extensions, most papers largely remain within the traditional VMI framework where replenishment decisions are optimized to minimize total cost, emphasizing cost-minimizing replenishment policies, with limited attention to how suppliers actively allocate constrained capacity among multiple buyers.

More recent studies have introduced industry-specific characteristics to better reflect real-world inventory dynamics. For instance, the authors develop a single-vendor multi-buyer VMI model with multiple quality-degraded raw materials in the agrifood industry and apply the whale optimization algorithm (WOA) to maximize total profit [16]. Although this work significantly enhances the practical relevance of the model, it mainly focuses on algorithmic performance and inventory dynamics, rather than providing structured decision support at the system level. In parallel, several studies have employed metaheuristic and fuzzy-based approaches to address large-scale and uncertain environments [17, 18]. However, these contributions generally lack an explicitly defined Decision Support System (DSS) architecture that links optimization outputs to managerial decision processes in semi-structured contexts.

Overall, the literature reveals three interrelated research gaps: the limited development of SMI optimization models in single-supplier multi-buyer settings, the underrepresentation of production capacity allocation as a central decision variable, and the lack of integration between optimization models and DSS frameworks. To address these gaps, this study develops a MILP-based DSS for the SS-MB-SMI problem, enabling scenario analysis and supporting managerial evaluation of trade-offs among capacity, service level, and inventory performance.

Table 1. Summarization of the existing approaches

Article	Problem characteristics							Solution Method	
	Industry*	Replenishment Programs**	Multiple prod-	Production Capacity	Order multiple	Overstock penalty	Stockout re-sponse***	Modelling and/or Algo-rithm****	DSS
[11]	_	V	_	_	X	_	_	NLP, SS	_
[12]	_	V	_	_	X	X	_	NLP, EXA	_
[13]	_	V	X	X	X	_	B & L	NLP, PSO, GA	_
[14]	_	V	_	X	X	X	B	NLP, EXA	_
[15]	_	V	X	X	_	_	B & L	NLP, LR	_

[16]	F	V	_	X	_	_	_	NLP, WOA	_
[17]	_	V	_	_	_	_	L	Fuzzy	_
[18]	_	V	_	X	_	_	B & L	NLP, PSO	_
[19]	F	V	_	X	_	_	_	NLP, EXA	_
[20]	_	V	_	_	_	X	_	NLP, HEU	_
[21]	F	V	_	X	_	_	L	GA	Y
[22]	_	V	_	_	X	_	_	GA	_
Our approach	A	S	X	X	X	X	B	MILP	Y

* A: Automotive, F: Food

** V: VMI, S: SMI

*** B: Backorder, L: Lost sales

**** EXA: Exact approaches, GA: Genetic Algorithm, HEU: Heuristic, NLP: Non-linear Programming, MILP: Mixed Integer Linear Programming, LR: Lagrangian Relaxation, PSO: Particle Swarm Optimization, SSM: Sequential Solution, WOA: Whale Optimization Algorithm

3 Problem Description and Formulation

3.1 Problem background

The considered SS-MB-SMI problem in this study is a vertically integrated supply chain where a single upstream component supplier (CS) serves multiple downstream finished-goods plants (FGPs) over a multi-period horizon (Fig. 1). Each plant requires specific component quantities, and shipments must comply with predefined case-pack requirements for logistics efficiency.

The supplier operates under limited capacity, which must be allocated across plants in each period. Inventory at each plant fluctuates over time based on incoming shipments and demand fluctuations, starting from an initial level. To both ensure stability and reduce the risk of supply chain uncertainty, inventory is controlled within predefined upper and lower bounds.

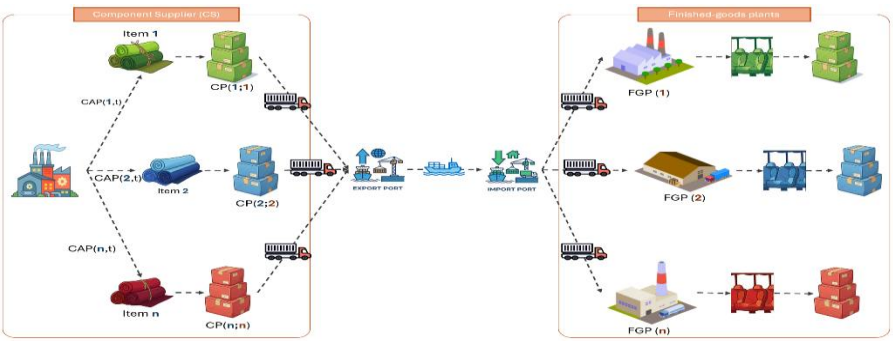

Fig. 1. The component supply chain

3.2 Mathematical Model

The SS-MB-SMI problem is formulated as a MILP model to allocate supplier capacity across FGPs while maintaining inventory within predefined limits. The model serves as the analytical core of the DSS, generating optimal allocation decisions for scenario evaluation and managerial support.

Asumption

The following assumptions are adopted to ensure model tractability while maintaining practical relevance:

1. All input parameters are deterministic.
2. Stockouts are allowed and can be fulfilled as backorders.
3. The supplier's production capacity for each item is predetermined and fixed for each period. Consequently, the total quantity of firmed purchase orders in each period must equal the available capacity.
4. The shipment lead time from the CS to the FGPs is constant. Because the supplier prepares inventory in advance to cover this fixed lead time, the transportation delay is not explicitly modeled.

Indices, parameters, and decision variables

The indices, parameters, and decision variables used in the model are summarized in Appendix.

Objective Function

The objective is to minimize the total operational cost across all items, plants, and planning periods.

$$Min\left[\sum_i^I \sum_j^J \sum_t^T \left(C_{i,j,t}^o \times o_{i,j,t} + C_{i,j,t}^s \times s_{i,j,t} + C_{i,j,t}^b \times bo_{i,j,t} + C_{i,j,t}^p \times p_{i,j,t}\right)\right] \quad (1)$$

The objective function consists of the following cost components:

- Backorder cost: arise when components are unavailable at FGPs, causing production shortages and delaying manufacturing to a later period.
- Overstock cost: occur when net inventory at FGPs exceeds the inventory ceiling $U_{i,j,t}$, including extra warehouse space, handling, and potential obsolescence costs.
- Shortage cost: reflects the impact of net inventory falling below the inventory floor $L_{i,j,t}$, increasing the risk of stockouts under demand uncertainty.
- Penalty cost: result from shipments that do not meet case-pack requirements, causing inefficient box utilization

Constraints

Inventory Balance
The inventory dynamics at each FGP are governed by the following balance equations.

For the first period:

$$I_{i,j,1} = BI_{i,j} + \Delta I_{i,j,1} + q_{i,j,1} \times CP_{i,j} + r_{i,j,1},\ \forall i, j \tag{2}$$

For subsequent periods:

$$I_{i,j,t} = I_{i,j,(t-1)} + \Delta I_{i,j,t} + q_{i,j,t} \times CP_{i,j} + r_{i,j,t},\ \forall i, j, t \geq 2 \tag{3}$$

These constraints track inventory evolution based on incoming shipments and demand-related fluctuations.

Supplier Capacity
The supplier production capacity for each item in each period must be fully allocated among all FGPs:

$$\sum_j^J \left(q_{i,j,t} \times CP_{i,j} + r_{i,j,t}\right) = CAP_{i,t},\ \forall i, t \tag{4}$$

This constraint ensures that the total allocated shipment quantity equals the available production capacity.

Inventory Deviation
Backorders, overstock, and shortage quantities are determined relative to the resulting inventory level.

Backorders:

$$bo_{i,j,t} \geq -I_{i,j,t},\ \forall i, j, t \tag{5}$$

Overstock:

$$o_{i,j,t} \geq I_{i,j,t} - U_{i,j,t},\ \forall i, j, t \tag{6}$$

Shortage relative to safety stock floor:

$$s_{i,j,t} \geq L_{i,j,t} - I_{i,j,t},\ \forall i, j, t \tag{7}$$

These variables quantify deviations from desired inventory ranges.

Case-Pack Violation Linearization
To capture inefficient packaging when shipment quantities are not divisible by the case-pack size, a binary penalty variable is introduced.

$$r_{i,j,t} \leq HV \times p_{i,j,t},\ \forall i, j, t \tag{8}$$

$$r_{i,j,t} \geq HV \times \left(p_{i,j,t} - 1\right) + 1,\ \forall i, j, t \tag{9}$$

These constraints activate the penalty variable when residual units occur.

Variable Domains

$$bo_{i,j,t}, o_{i,j,t}, s_{i,j,t} \geq 0, \ \forall i, j, t \tag{10}$$

$$q_{i,j,t}, r_{i,j,t} \in N, \ \forall i, j, t \tag{11}$$

$$p_{i,j,t} \in \{0; 1\}, \ \forall i, j, t \tag{12}$$

4 DSS Framework for MILP-Based SMI Coordination

The Decision Support System (DSS) developed in this study operationalizes the SMI coordination problem described in Section 3 and implements the MILP formulation presented in Section 4. In this context, the DSS supports semi-structured decision problems in which part of the decision process can be formally modeled and optimized, while the remaining part requires managerial judgment.

Rather than replacing the planner, the DSS assists decision makers by integrating optimization-based allocation decisions with interactive evaluation and scenario analysis. Through this integration, the system enables planners to evaluate alloca-tion policies under different operational conditions while maintaining human oversight in the final decision process [23].

4.1 System architecture

The system architecture integrates enterprise data, optimization modeling, and managerial evaluation within a unified, layered structure as illustrated in Fig. 2. Rather than operating as a standalone optimization engine, the DSS connects data processing, model execution, and user interaction in a continuous decision loop.

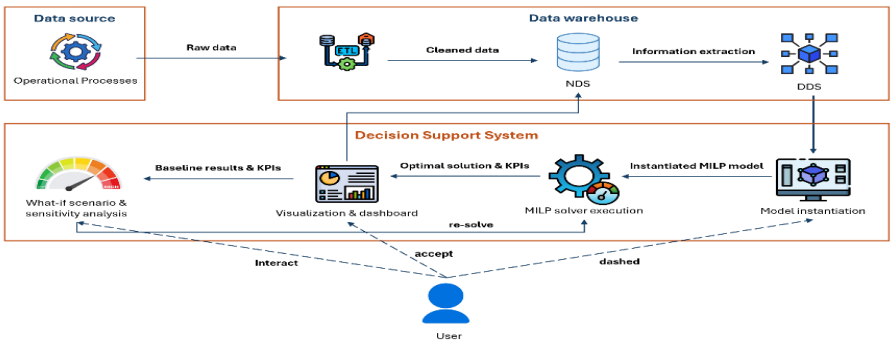

Fig. 2. DSS architecture

Data Source Layer

The architecture begins with operational processes that generate raw transactional data related to production, inventory, and distribution. These data form the primary input to the analytical system.

Data Warehouse Layer
Raw data are processed through an ETL procedure and stored in the Normalized Data Store (NDS), where consistency, completeness, and temporal alignment are ensured. The NDS preserves detailed operational records. For analytical purposes, selected data are then extracted and reorganized into a Dimensional Data Store (DDS). The DDS structures information according to the core indices of the MILP model—items, FGPs, and time periods—thereby enabling direct mapping between stored data and model parameters.

Decision Support System Layer
Based on the DDS, the DSS performs model instantiation by translating structured data into a fully specified MILP model. Decision variables, constraints, and cost parameters are initialized in accordance with the mathematical formulation. The instantiated model is then solved using a MILP solver to generate optimal allocation decisions under capacity and inventory constraints. The resulting outputs, including decision variables and KPIs, are transferred to a visualization and dashboard module for managerial evaluation. The same structure also enables scenario adjustments, allowing planners to modify key parameters and re-execute the model to conduct what-if and sensitivity analyses.

User Interaction and Feedback Loop
Through the dashboard interface, decision makers can review baseline results, conduct what-if and sensitivity analyses, and either accept the solution or trigger re-solving with adjusted inputs. This establishes a controlled feedback loop linking user judgment with model re-execution, without bypassing the underlying optimization logic.

5 Experiment Evaluation

5.1 Industrial Context and Case Study Description

The empirical evaluation is based on a real-world case study from a global automotive manufacturer operating under a SMI structure. In this context, a Vietnam-based CS produces seat trim cover kits delivered to six US-based FGPs, covering a portfolio of 943 distinct items over a four-week planning horizon. Through real-time information sharing, supplier-level replenishment decisions are aligned with downstream demand and capacity constraints.

The supplier plans production several weeks ahead based on rolling forecasts from FGPs. Shipment lead time is assumed constant and omitted, allowing allocation decisions to align closely with demand realization. This configuration reflects the

complexity of coordinating supplier-driven replenishment across multiple products and geographically dispersed plants.

5.2 Experimental Design and Test Scenarios

Three representative scenarios are constructed to evaluate the proposed model and DSS, each with three warehouses and four planning periods under common baseline parameters (Table 2), including: (i) backorder-dominated imbalance, (ii) overstock risk, and (iii) capacity indivisibility under case-pack constraints. As a result, these scenarios assess allocation behavior under different structural conditions affecting supply–demand balance and inventory policies.

Table 2. Baseline parameters

Inventory floor	2	Overstock unit cost	2
Inventory ceiling	25	Shortage unit cost	5
Backorder unit cost	200	Penalty cost	100

Scenario 1: Backorder-dominated imbalance

The first scenario examines allocation under severe demand imbalance, where a large demand shock at WH01 leads to significant backorders under limited capacity (Table 3). The result indicated that the model prioritizes allocating capacity to WH01 in early periods to reduce backorders to reflect the dominance of backorder cost. Consequently, warehouses with higher shortage pressure receive priority in order allocation.

Table 3. Test scenario 1: Backorder existance

Capacity = 12			Period			
Warehouse	Data Type	Case-pack	1	2	3	4
WH01	Beginning inventory	4	10			
	Inventory fluctuation	4	-40	-2	-2	-2
	Net inventory (before)	4	-30	-32	-34	-36
	Pack quantity	4	3	3	3	1
	Residual	4	0	0	0	0
	Net inventory (after)	4	-18	-8	2	4
WH02	Beginning inventory	4	15			
	Inventory fluctuation	4	-3	-3	-3	-3
	Net inventory (before)	4	12	9	6	3
	Pack quantity	4	0	0	0	1
	Residual	4	0	0	0	0
	Net inventory (after)	4	12	9	6	7
WH03	Beginning inventory	4	12			
	Inventory fluctuation	4	-3	-3	-3	-3
	Net inventory (before)	4	9	6	3	0
	Pack quantity	4	0	0	0	1

	Residual	4	0	0	0	0
	Net inventory (after)	4	9	6	3	4

Scenario 2: Overstock risk

The second scenario investigates allocation under persistent overstock conditions, where WH01 and WH02 exceed inventory ceilings (Table 4). The model allocates all capacity to WH03, avoiding additional supply to overstocked warehouses due to their sustained inventory surplus. This result shows that allocation decisions are driven by inventory bound violations and cost structures, prioritizing shortage mitigation without increasing overstock.

Table 4. Test scenario 2: Overstock penalty

Capacity = 8			Period			
Warehouse	Data Type	Case-pack	1	2	3	4
WH01	Beginning inventory	4	35			
	Inventory fluctuation	4	0	0	0	0
	Net inventory (before)	4	35	35	35	35
	Pack quantity	4	0	0	0	0
	Residual	4	0	0	0	0
	Net inventory (after)	4	35	35	35	35
WH02	Beginning inventory	4	32			
	Inventory fluctuation	4	0	0	0	0
	Net inventory (before)	4	32	32	32	32
	Pack quantity	4	0	0	0	0
	Residual	4	0	0	0	0
	Net inventory (after)	4	32	32	32	32
WH03	Beginning inventory	4	12			
	Inventory fluctuation	4	-5	-5	-5	-5
	Net inventory (before)	4	7	2	-7	-12
	Pack quantity	4	2	2	2	2
	Residual	4	0	0	0	0
	Net inventory (after)	4	15	18	21	24

Scenario 3: Capacity indivisibility under case-pack constraints

The third scenario analyzes the impact of indivisible case-pack sizes, where capacity is not divisible by the case-pack, creating residual units. For example, the production capacity is set to 13 units per period, which is not divisible by the case-pack size of 4, generating residual quantities (Table 5). The result shows that the model assigns the residual to a single warehouse while maintaining full case-pack allocations elsewhere, thereby concentrating and minimizing penalty costs while ensuring feasibility.

Table 5. Non-optimal case-pack allocation

Capacity = 13			Period			
Warehouse	Data Type	Case-pack	1	2	3	4
WH01	Beginning inventory	4	10			
	Inventory fluctuation	4	-4	-4	-4	-4
	Net inventory (before)	4	6	2	-2	-6
	Pack quantity	4	0	3	1	3
	Residual	4	1	1	0	0
	Net inventory (after)	4	7	16	16	24
WH02	Beginning inventory	4	10			
	Inventory fluctuation	4	-4	-4	-4	-4
	Net inventory (before)	4	6	2	-2	-6
	Pack quantity	4	0	0	2	0
	Residual	4	0	0	0	1
	Net inventory (after)	4	6	2	6	3
WH03	Beginning inventory	4	10			
	Inventory fluctuation	4	-4	-4	-4	-4
	Net inventory (before)	4	6	2	-2	-6
	Pack quantity	4	3	0	0	0
	Residual	4	0	0	1	0
	Net inventory (after)	4	18	14	11	7

5.3 Computation Performance and Optimization Results

Computational Scalability

The MILP model is implemented in Python using PuLP and solved with the CBC solver. Across 943 item-level subproblems, the average solving time is approximately 0.05 seconds, with a minimum of 0.01 seconds and a maximum of 0.66 seconds. As illustrated in Fig. 3, the entire batch is completed within 52 seconds. These results indicate that the proposed framework can be integrated into routine operational planning without introducing computational bottlenecks.

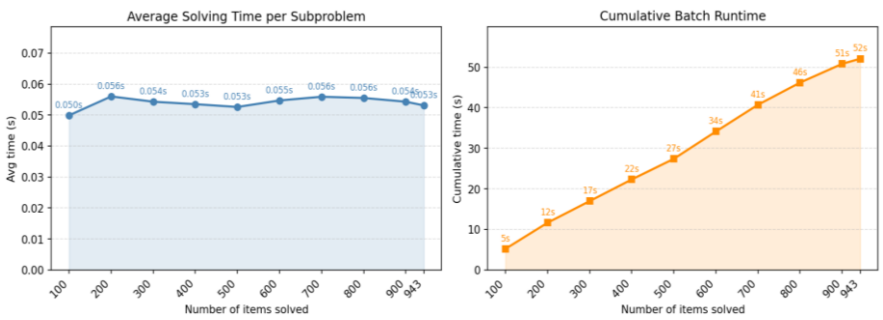

Fig. 3. Computational scalability of the proposed MILP framework

Baseline Cost Structure

Under the baseline scenario, the total system cost amounts to 6,203,728. The cost breakdown reveals that backlog costs dominate the overall system cost, while overstock and shortage costs contribute only marginally. This indicates that service failures, rather than inventory holding, represent the primary economic driver in the system.

Such a cost structure is common in service-oriented distribution networks where unmet demand generates a significantly higher economic impact than holding additional inventory. Consequently, reducing backlog becomes essential for maintaining operational efficiency and service reliability.

Impact of Demand and Capacity Variations

To evaluate the robustness of the proposed framework under different operational conditions, two types of scenario analyses are conducted: demand variations and reductions in supply capacity. These situations frequently occur in real-world supply chains due to sudden demand surges or disruptions in production and transportation capacity.

Table 6 presents the results for demand scenarios ranging from −50% to +50% relative to the baseline level. The results show that total system cost increases rapidly as demand rises. When demand increases by 50%, the total cost grows from 6,203,728 to 24,833,617.7, whereas a 50% demand reduction lowers the cost to 191,800.95. This increase is largely driven by backlog growth. In fact, backlog costs consistently account for the vast majority of total cost across all demand scenarios, indicating that shortages dominate the economic behavior of the system. As demand exceeds the effective supply capability, unmet demand accumulates quickly and escalates operational costs. At the same time, overstock costs decrease under higher demand conditions, reflecting the gradual absorption of excess inventory.

Table 6. Cost components under different demand scenarios

Demand	Total	Backlog	Overstock	Shortage	Penalty
−50%	191,801	75,000	103,721	13,080	0
−20%	2,453,519	2,342,700	89,965	20,854	0
Baseline	6,203,728	6,093,000	81,621	29,107	0
+20%	12,473,965	12,360,600	73,898	39,467	0
+50%	24,833,618	24,709,500	63,861	60,257	0

Table 7 reports the results when supply capacity is reduced. As capacity declines, total system cost rises significantly, reaching 24,927,727.9 when capacity is reduced by half. Similar to the demand scenarios, backlog costs explain most of the increase. When supply capacity becomes insufficient, shortages propagate through the network and rapidly escalate operational costs, indicating that capacity limitations can amplify service-level failures in the distribution system. In addition, penalty costs appear under

severe capacity reductions, reflecting situations where certain operational constraints must be relaxed to maintain feasible allocation decisions.

Table 7. Cost components under reduced supply capacity scenarios

Capacity	Total	Backlog	Overstock	Shortage	Penalty
−10%	8,369,714	8,204,400	73,320	33,995	58,000
−30%	13,966,219	13,803,900	57,712	46,608	58,000
−50%	24,927,728	23,166,000	44,671	67,057	1,650,000

5.4 Managerial Implications

The results provide several managerial insights. First, backlog costs dominate the system cost across most scenarios, suggesting that preventing severe shortages should take priority over minimizing inventory holding costs. Second, the system is highly sensitive to demand variability, highlighting the importance of accurate demand forecasting and effective information sharing across the supply chain. Finally, supply capacity acts as a key structural constraint in the network, indicating that maintaining flexible capacity or modest safety buffers can help stabilize operations and reduce service disruptions.

6 Conclusion

This study addresses the SMI problem in a vertically integrated supply chain by developing a MILP model that optimally allocates production capacity across multiple FGPs under inventory bounds, case-pack constraints, and cost considerations. The model is embedded within a DSS to support practical decision-making through integrated data processing, optimization, and visualization. Empirical results from a real automotive case demonstrate that the framework is computationally efficient and achieves significant cost reductions compared to baseline policies, while effectively adapting to varying operational conditions.

Future research may extend the current framework by incorporating demand uncertainty, stochastic production capacity, and dynamic lead-time considerations. In addition, integrating real-time data streams and advanced forecasting methods into the DSS could further enhance its applicability in digital supply chain environments.

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Appendix

Table 8. Indices, parameters, and decision variables

	Symbol	Description	Symbol	Description
Indi-ces	i	Items, $i = (1,2,\dots,I)$		
	j	Finished-goods plants, $j = (1,2,\dots,J)$	t	Time periods, $t = (1,2,\dots,T)$
parameters	$BI_{i,j}$	Beginning inventory	$CP_{i,j}$	Case-pack size
	$C^b_{i,j,t}$	Backorder unit cost	$C^o_{i,j,t}$	Overstock unit cost
	$C^s_{i,j,t}$	Shortage unit cost	$C^p_{i,j,t}$	Penalty cost
	$U_{i,j,t}$	Inventory ceiling	$L_{i,j,t}$	Inventory floor
	$CAP_{i,t}$	Production capacity	$\Delta I_{i,j,t}$	Inventory fluctuation
	HV	Large constant		
variables	$q_{i,j,t}$	Number of full case-packs	$r_{i,j,t}$	Residual units
	$I_{i,j,t}$	Net inventory	$bo_{i,j,t}$	Backorder units
	$s_{i,j,t}$	Shortage units	$o_{i,j,t}$	Overstock units
	$p_{i,j,t}$	Binary variable indicating case-pack violation		